

Undergraduate Texts in Mathematics

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Introduction to Linear Algebra

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The method of changing a matrix by row equivalences to put it in row echelon form works in general.

Theorem 4.1. *Every matrix is row equivalent to a matrix in row echelon form.*

Proof. Select a non-zero entry furthest to the left in the matrix. If this entry is not in the first column, this means that the matrix consists entirely of zeros to the left of this entry, and we can forget about them. So suppose this non-zero entry is in the first column. After an interchange of rows, we can find an equivalent matrix such that the upper left-hand corner is not 0. Say the matrix is

$$\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

and $a_{11} \neq 0$. We multiply the first row by a_{21}/a_{11} and subtract from the second row. Similarly, we multiply the first row by a_{i1}/a_{11} and subtract it from the i -th row. Then we obtain a matrix which has zeros in the first column except for a_{11} . Thus the original matrix is row equivalent to a matrix of the form

$$\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ 0 & a'_{22} & \cdots & a'_{2n} \\ \vdots & \vdots & & \vdots \\ 0 & a'_{m2} & \cdots & a'_{mn} \end{pmatrix}.$$

We then repeat the procedure with the smaller matrix

$$\begin{pmatrix} a'_{22} & \cdots & a'_{2n} \\ \vdots & & \vdots \\ a'_{m2} & \cdots & a'_{mn} \end{pmatrix}.$$

We can continue until the matrix is in row echelon form (formally by induction). This concludes the proof.

Observe that the proof is just another way of formulating the elimination argument of §3.

We give another proof of the fundamental theorem:

Theorem 4.2. *Let*

$$\begin{aligned} a_{11}x_1 + \cdots + a_{1n}x_n &= 0, \\ \vdots & \\ a_{m1}x_1 + \cdots + a_{mn}x_n &= 0, \end{aligned}$$